$$(\underline{A}\times\overbrace{\underline{B}\times\underline{C}}^{D})_i = (\underline{A}\times\underline{D})_i = e_{ijk}A_jD_k = e_{ijk}A_j e_{kmn}B_mC_n =$$

$$(\underline{D})_k = (\underline{B}\times\underline{C})_k = e_{kmn}B_mC_n$$

$$(\underline{A}\times\underline{B}\times\underline{C})_i = e_{ijk}e_{kmn}A_jB_mC_n = e_{kij}e_{kmn}A_jB_mC_n =$$

$$= (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})A_jB_mC_n$$

$$= A_jB_iC_j - A_jB_jC_i$$

$$= -e_{kji}\,e_{kmn}$$

$$= [(\underline{A}\cdot\underline{C})\underline{B}]_i - [(\underline{A}\cdot\underline{B})\underline{C}]_i = [(\underline{A}\cdot\underline{C})\underline{B} - (\underline{A}\cdot\underline{B})\underline{C}]_i$$

$$\underline{A}\times\underline{B}\times\underline{C} = (\underline{A}\cdot\underline{C})\underline{B} - (\underline{A}\cdot\underline{B})\underline{C}$$